



Figure 1: FreeNAS

Features

- Automatic failover without a centralised metadata server
- Layered approach to the file system, where features are added or removed on demand
- Disk file systems such as ext3, ext4, xfs, etc., are supported to store the data
- Can scale up to petabytes of storage
- Distributed geo-replication
- File snapshots
- Compression translator

FreeNAS: A NAS/SAN solution

FreeNAS is a FreeBSD-based operating system for network attached storage. It comes with an easy-to-use Web UI. NAS-oriented open source storage distributions of Linux and FreeBSD include FreeNAS, Gluster, Openfiler, etc., which can be connected and configured using a Web browser. It is released under a BSD licence. The FreeNAS Project was initially founded by Olivier Cochard-Labbe in 2005. It was PHP-based, easy to use and was based on an embedded firewall (also based on FreeBSD). FreeNAS can run from a USB disk or virtual machine. It is distributed as an ISO image. FreeNAS is available for both 32-bit and 64-bit architectures (a 32-bit system can only address up to 4 GB of RAM). The FreeNAS stable version is 9.2.1.5.

Features

- Supports protocols such as NFS, SSH, FTP, TFTP, BitTorrent, iTunes, CIFS (via Samba), etc
- Plug-in support for SlimServer and Xbox Media Stream Protocol
- iSCSI target feature is used to create virtual disks. ZFS, UFS, ext2, and ext3 are also supported.
- Read and write for FAT32 and NTFS
- Hard drives supported are (P/S)-ATA, SCSI, iSCSI, USB and FireWire; and booting is possible from a HDD, CD-ROM, floppy disk or USB flash drive.



Figure 2: Openfiler

- It provides snapshot support, deduplication, a removable log device, etc.

Openfiler: A NAS/SAN solution

Openfiler is a free network storage OS created by Xinit Systems, and is based on the rPath Linux distribution. It supports file-based Network Attached Storage (NAS) and block-based Storage Area Networking (SAN) functionality in a single unified framework. Openfiler uses the Linux 2.6 kernel base to deliver a wide-ranging storage management solution that fulfils the needs of enterprise applications. Openfiler supports on-disk file systems such as XFS, ext3, ReiserFS v3 and JFS. XFS and ext3 are journalled file systems that enhance data security and reduce the need for regular file system checks. Openfiler has the following hardware requirements: an x86- or x64-based computer with at least a 64-bit processor with 1.6 GHz or more, 512 MB of RAM, a CD-ROM or DVD-ROM drive if you are performing a local install, 10 GB of hard disk space (8 GB for OS installation and 2 GB for swap space). It is licensed under the GNU General Public License version 2. It supports the i386/AMD64 platforms. Installable images are available for x86_64 architectures. The download options are available at <http://www.openfiler.com/community/download>

Features

Full industry-standard protocol suite: NFSv3 support for all UNIX-based clients with additional support for ACL protocol extensions; CIFS or SMB support for Microsoft Windows-based clients; FTP support, WebDAV and HTTP 1.1 support; and NFSv4 support (testing)

Good block storage virtualisation: Point-in-time snapshots support with scheduling; synchronous/asynchronous volume replication and migration; optimal division of storage by full iSCSI target support. It also supports virtual iSCSI targets.

Extensive share management: Support for multiple